

Convex Illumination Problem (1 pt)

Show that the solution $p^* = (p_1^*, p_2^*, \dots, p_n^*)^T \in \mathbb{R}^n$ of the non-convex illumination problem from the lecture

$$\begin{aligned} & \text{minimize} && \max_{k=1 \dots m} |\log I_k - \log I_{des}| \\ & \text{subject to} && 0 \leq p_j \leq p_{max}, \quad j = 1 \dots n \end{aligned}$$

with $I_k = \sum_{j=1}^n a_{kj} p_j$ for geometric constants $a_{jk} \in \mathbb{R}$, a constant desired illumination $I_{des} \in \mathbb{R}$ and an upper bound $p_{max} \in \mathbb{R}$ on the lamp power, is identical to the solution of the following equivalent (convex) problem

$$\begin{aligned} & \text{minimize} && \max_{k=1 \dots m} h(I_k / I_{des}) \\ & \text{subject to} && 0 \leq p_j \leq p_{max}, \quad j = 1 \dots n \end{aligned}$$

with $h(u) = \max\{u, 1/u\}$.

The objective function can be rewritten:

$$\max_{k=1 \dots m} |\log I_k - \log I_{des}| = \max_{k=1 \dots m} \left| \log \frac{I_k}{I_{des}} \right|$$

We define $u_k = \frac{I_k}{I_{des}}$. We know that $|\log u| = \max\{\log u, -\log u\}$ for $u > 0$.
 $-\log u = \log \frac{1}{u}$.

$\log(x)$ is monotone increasing for $x \in (0, \infty)$

$$\text{so } \max\{\log u, -\log u\} = \max\{\log u, \log \frac{1}{u}\} = \log \max\{u, \frac{1}{u}\} = \log h(u)$$

Applying this to the objective:

$$\max_{k=1 \dots m} |\log I_k - \log I_{des}| = \max_{k=1 \dots m} \left| \log \frac{I_k}{I_{des}} \right| = \max_{k=1 \dots m} \log h(u) = \log \max_{k=1 \dots m} h(u)$$

As the $\log$ is monotone increasing, minimizing $\log \max_{k=1 \dots m} h(u)$ is the same as minimizing $\max_{k=1 \dots m} h\left(\frac{I_k}{I_{des}}\right)$ //